國立高雄大學 109 學年度轉學招生考試試題(轉二年級)

科目：微積分
考試時間：80 分鐘

系所：
電機工程學系(無組別)
本科原始成績：100 分

是否使用計算機：是

■ 以下考題總共十題，每題為十分。請根據題號依序作答。

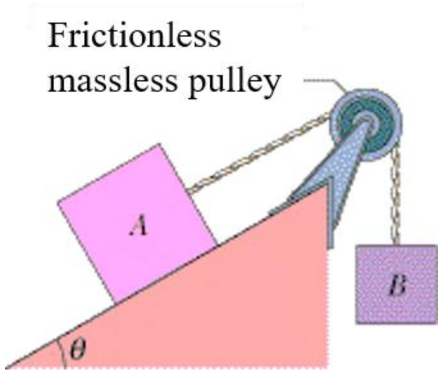
1. Find $\lim_{x \to 2}(x^2+3)$.
2. Find $\frac{d}{dx}[\csc 4x(1+\tan 2x)]$.
3. $x\cos(\sin y)=0$. Determine $\frac{dy}{dx}$.
4. Find $\int e^x(e^x+1)^{50}dx$.
5. Find $\frac{d}{dx}[3^{5^x}]$.
6. Find $\int \frac{e^x}{e^{2x}+25}dx$.
7. Find $\int_0^{\frac{\pi}{2}}\left(\frac{\sin x}{0.5-\cos x}\right)dx$.
8. $w=xy+\tanh^{-1}z+yu$. Determine the total differential of w.
9. $\vec{u}=\cos\theta\vec{\imath}+\sin\theta\vec{\jmath}$ and $\nabla f(x,y)=x\vec{\imath}+y\vec{\jmath}$. Find $D_{\vec{u}}f(x,y)$.
10. Find $\int_1^2\int_1^2\int_1^2(x+y+z)dzdydx$.

國立高雄大學 109 學年度轉學招生考試試題(轉二年級)

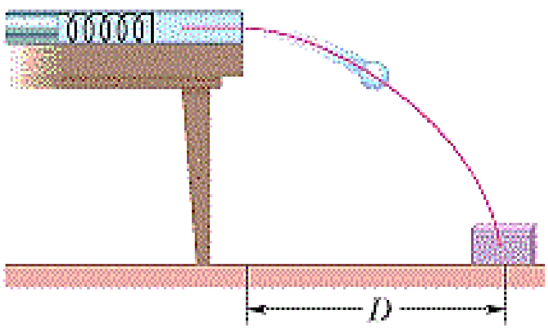
科目：物理　　系所：電機工程學系(無組別)　　是否使用計算機：是
考試時間：80 分鐘　　本科原始成績：100 分

*Gravitational acceleration, $g = 9.8$ m/s^2

1. (10%) In the figure, two blocks are connected over a pulley. The mass of block A is 10 kg and the coefficient of kinetic friction between *A* and the incline is 0.2. Angle θ of the incline is 30°. Block *A* slides down the incline at constant speed. What is the mass of block *B*? (Note: sin30º = 0.5, cos30º = 0.866)



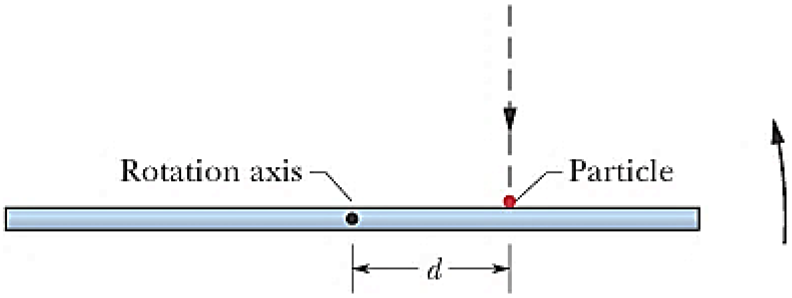
2. (10%) In the figure, two children (Bobby and Rhoda) are playing a game in which they try to hit a small box on the floor with a marble fired from a spring-loaded gun that is mounted on a table. The target box is horizontal distance D = 2.2 m from the edge of the table. Bobby compresses the spring 1.1 cm, but the center of the marble falls 26 cm short of the center of the box. How far should Rhoda compress the spring to score a direct hit? (Assume that neither the spring nor the ball encounters friction in the gun.)



3. (10% + 5%) The figure is an overhead view of a thin uniform rod of length (L) 0.6 m and mass M rotating horizontally at 80 rad/s counterclockwise about an axis through its center. A particle of mass M/3 and traveling horizontally at speed 40 m/s hits the rod and sticks. The particle's path is perpendicular to the rod at the instant of the hit, at a distance *d* from the rod's center. (Note: the rotational inertia for thin rod about axis through center perpendicular to length: $ML^2/12$)

(a). At what value of *d* are rod and particle stationary after the hit?

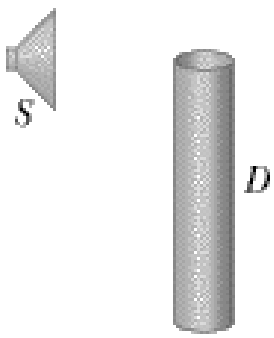
(b). In which direction (counterclockwise or clockwise) do rod and particle rotate if *d* is greater than the value in question (a)? (Please briefly explain the reason)



4. (5% + 5%) In the figure, *S* is a small loudspeaker driven by an audio oscillator and amplifier, adjustable in frequency from 660 Hz to 2000 Hz only. Tube *D* is a piece of cylindrical sheet-metal pipe 59.7 cm long and open at both ends.

(a). If the speed of sound in air is 344 m/s at the existing temperature, at how many frequencies will a resonance occur in the pipe when the frequency emitted by the speaker is varied from 660 Hz to 2000 Hz?

(b). What is the lowest resonant frequency in the given frequency interval (660 Hz to 2000 Hz)?



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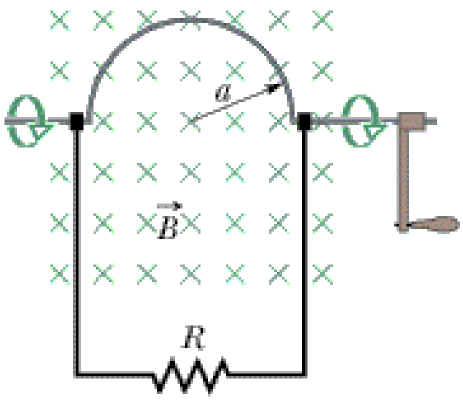
5. (5% + 5%) In the figure, a stiff wire bent into a semicircle of radius a = 3.5 cm is rotated at constant angular speed 40 rev/s in a uniform **B** = 28 mT magnetic field (direction: into the page). What are the (a) frequency and (b) the amplitude of the emf induced in volts in the loop?

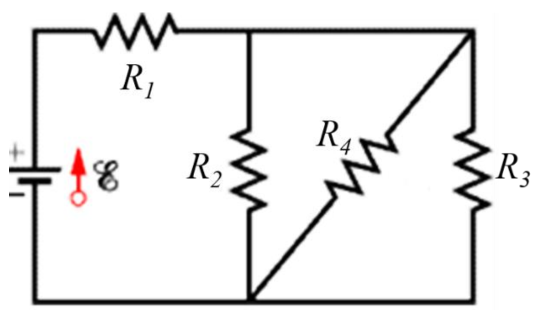


6. (5% + 5% + 5%) In the circuit of figure, R_1 = 100 Ω, R_2 = R_3 = 50 Ω, R_4 = 75 Ω, and the ideal battery has emf ε = 12 V.

(a). What is the equivalent resistance of this circuit?

(b). What is current in resistance R_1?

(c). What is current in resistance R_4?

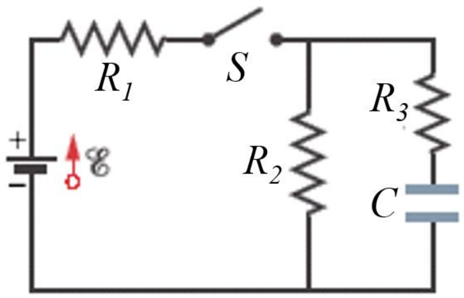


7. (5% + 5% + 10%) In the circuit of the figure, an emf ε = 1.2 kV, C = 6.5 μF, $R_1 = R_2 = R_3$ = 0.8 MΩ. With capacitor completely uncharged, switch S is suddenly closed at time (t = 0).

At t = 0, (a) what is the current in resistor R_2?

At t → ∞, (b) what are the current in resistors R_2 and R_3?

What is the potential difference V_2 across resistor R_2 at (c) t = 0 and (d) t = ∞?



8. (10%) A typical "light dimmer" used to dim the stage lights in a theater consists of a variable inductor L (whose inductance is adjustable between zero and L_{max}) connected in series with a lightbulb B as shown in the figure. The electrical supply is 120 V (rms) at 60.0 Hz; the lightbulb is rated as 120 V, 1300 W. What L_{max} is required if the rate of energy dissipation in the lightbulb is to be varied by a factor of 6 from its upper limit of 1300 W? Assume that the resistance of the lightbulb is independent of its temperature.